



Lab 11

VGA Controller

Name:	ID:	Section:
-------	-----	----------

11.1. Objectives

After performing this lab, student should be able to:

- Get familiarize with the VGA display and the required modules to design VGA controller.
- Develop a code for VGA synchronization circuit to generate *h_sync* and *v_sync* signals
- Develop a code for the pixel generation circuit. This circuit will determine the pixel to show the desired image in VGA Monitor
- Combine all the defined Verilog modules together to get the system working and test its functionality.
- Modify pixel generation module to display different images on VGA monitor using Basys-3 Board

11.2. Introduction

VGA (Video Graphics Array) is a video display standard introduced in the late 1980s in IBM PCs and allowed for a display resolution of 640x480 pixels it is widely supported by PC graphics hardware and monitors.

In previous Lab, Horizontal, vertical counter (*h_counter*, *v_counter*) and clock divider (*clock_div*) modules have already been developed. You have already developed horizontal and vertical counter (*h_counter*, *v_counter*) and clock divider (*clock_div*) modules in previous. In this lab, you are going to develop remaining modules of VGA Controller. A video controller generates the synchronization signals and outputs data pixels. A complete block diagram of a VGA controller is shown in Figure 11. 7. It contains an additional synchronization circuit, labeled *vga_sync*, and a pixel generation circuit labelled *pixel_gen*.

11.3. Working of VGA controller

A VGA monitor operates using an electron beam that scans the screen row by row, starting at the upper left corner and ending at the lower-right corner as shown in Figure 11. 1. This beam moves using two synchronization signals, called *h_sync* (horizontal synchronization), and *v_sync* (vertical synchronization). The *h_sync* signal tells the beam when to move to the next row. The *v_sync* signal tells the beam when to move to back to the top of the screen. To display a picture on the screen, we simply generate these synchronization signals and provide the pixel color to display on the screen.

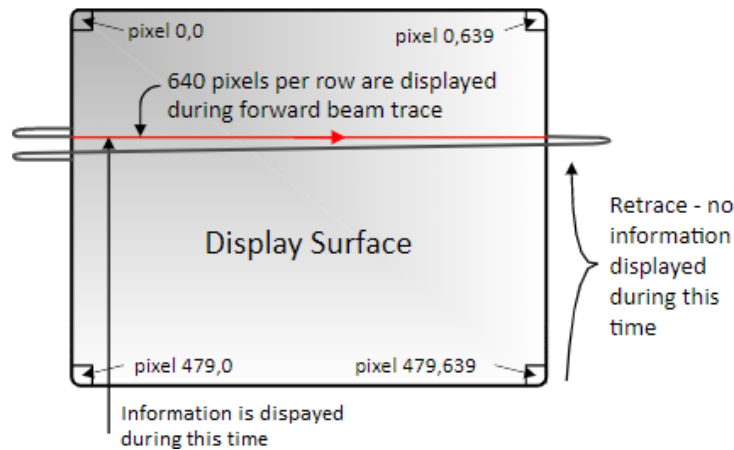


Figure 11. 1

11.3.1. Synchronization Circuit (*vga_sync*)

The *vga_sync* circuit generates timing and synchronization signals. It takes *h_count* and *v_count* as input and generate five outputs as shown in Figure 11. 2.

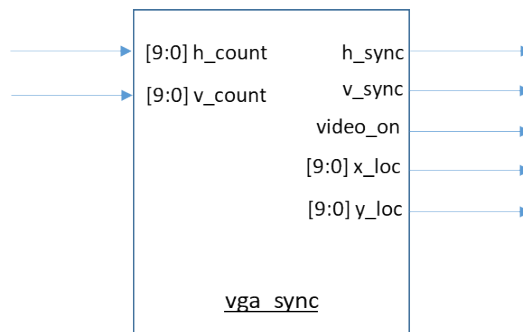


Figure 11. 2

- The *h_sync* and *v_sync* signals are connected to the VGA port to control the horizontal and vertical scans of the monitor in transverse manner. (details will be discussed in next heading)
- the *video_on* signal indicate whether the current targeted pixel is in the displayable region. It is asserted only when the *h_count* smaller than 640 and *v_count* is smaller than 480.
- The *x_loc* (the location of pixel x) *y_loc* (the location of pixel y) signals. The *x_loc* and *y_loc* specify the location of the current pixel and can be obtained from *h_count* and *v_count*.

The screen of a monitor includes a small black border, as shown at the top of Figure 11. 3. The middle rectangle is the visible portion. Note that the coordinate of the vertical axis increases downward. The coordinates of the top-left and bottom-right comers are (0,0) and (639,479), respectively.

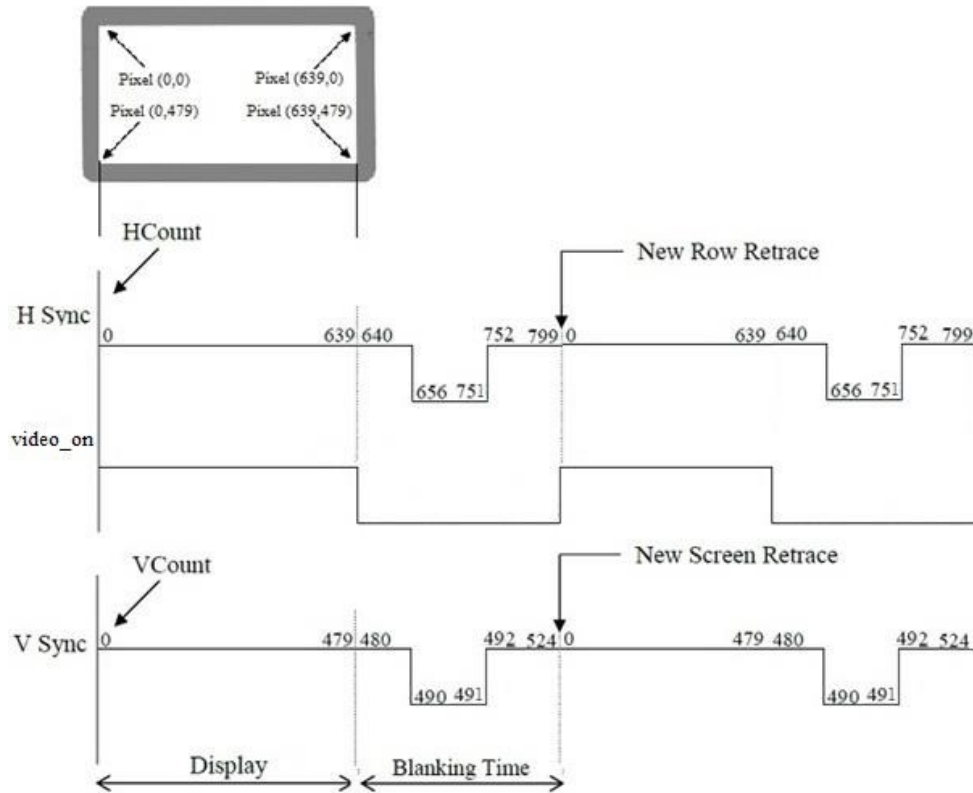


Figure 11. 3

a) Horizontal Synchronization (h_sync)

A detailed timing diagram of one horizontal scan is shown in Figure 11. 3. A period of the h_sync signal contains 800 pixels and can be divided into two regions:

- Display: region where the pixels are actually displayed on the screen. The length of this region is 640 pixels.
- Blanking: region has 3 segments
 - Retrace: region in which the electron beams return to the left edge. The video signal should be disabled (i.e., black), and the length of this region is 96 pixels.
 - Right border: region that forms the right border of the display region. It is also known as the front porch (i.e., porch before retrace). The video signal should be disabled, and the length of this region is 16 pixels.
 - Left border: region that forms the left border of the display region. It is also known as the back porch (i.e., porch after retrace). The video signal should be disabled, and the length of this region is 48 pixels.

Note that the lengths of the right and left borders may vary for different brands of monitors. The h_sync signal is obtained by $h_counter$ (mod-800 counter designed in previous lab). We intentionally start the counting from the beginning of the display region. This allows us to use

b) Vertical Synchronization

During the vertical scan, the electron beams move gradually from top to bottom and then return to the top. This corresponds to the time required to refresh the entire screen. The format of the *v_sync* signal is similar to that of the *h_sync* signal, as shown in Figure 11. 3. The time unit of the movement is represented in terms of horizontal scan lines. A period of the *v_sync* signal is 525 lines and can be divided into two regions:

- Display: region where the horizontal lines are actually displayed on the screen. The length of this region is 480 lines.
- Blanking: region has 3 segments
 - Retrace: region that the electron beams return to the top of the screen. The video signal should be disabled, and the length of this region is 2 lines.
 - Bottom border: region that forms the bottom border of the display region. It is also known as the front porch (i.e., porch before retrace). The video signal should be disabled, and the length of this region is 10 lines.
 - Top border: region that forms the top border of the display region. It is also known as the back porch (i.e., porch after retrace). The video signal should be disabled, and the length of this region is 33 lines.

As in the horizontal scan, the lengths of the top and bottom borders may vary for different brands of monitors.

The *v_sync* signal can be obtained by *v_counter* (mod-525 counter designed in previous lab). Again, we intentionally start counting from the beginning of the display region. This allows us to use the counter output as the vertical (y-axis) coordinate. This output constitutes the pixel-y signal. The *v_sync* signal goes low when the line count is 490 and 491.

Task a) VGA_sync Module

Design a *vga_sync* module as mentioned in Figure 11. 2. Follow the timing diagram in Figure 11.3 to write logic for *h_sync*, *v_sync*, *video_on*, *x_loc* and *y_loc* outputs. A skeleton code for *vga_sync* is given in Figure 11. 4. Provide complete code for the *vga_sync* Module.

```
`timescale 1ns / 1ps
module vga_sync (
input [9:0] h_count,
input [9:0] v_count,
output h_sync,
output v_sync,
output video_on,          // active area
output [9:0] x_loc,        // current pixel x- location
output [9:0] y_loc        // current pixel y-location
);

// horizontal
localparam HD =           // Horizontal Display Area
localparam HF =           // Horizontal (Front Porch) Right Border
localparam HB =           // Horizontal (Back Porch) Left Border
```

```

localparam HR =           // Horizontal Retrace

// vertical
localparam VD =           // Vertical Display Area
localparam VF =           // Vertical (Front Porch) BottomBorder
localparam VB =           // Vertical (Back Porch)Top Border
localparam VR =           // Vertical Retrace

.
.
.
.
.           //add your code here
.           //Note That the code written above is
.           // incomplete

endmodule           // vga_sync

```

Figure 11. 4



Verilog HDL local parameters are identical to parameters except that they cannot directly be modified by defparam statements or module instance parameter value assignments. Local parameters can be assigned constant expressions containing parameters, which can be modified with defparam statements or module instance parameter value assignments.

11.3.2. Pixel Generation Circuit

The pixel generation circuit generates the three 4-bit color signals, which are collectively referred to as the RGB signal. A color value is obtained according to the current coordinates of the pixel (the x_loc and y_loc signals) and the external control and data signals.

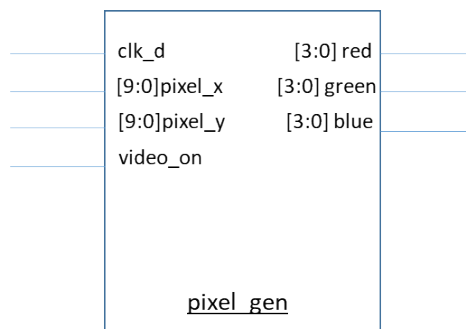


Figure 11. 5

Task b) *pixel_gen* Module

Write down the pixel generator module shown in the Figure 11. 6. This module has one input `clk_d` (of 25MHz), two 10-bit input `pixel_x` and `pixel_y` and one input `video_on`. This module generates three 4-bit signals of red, green and blue color.

```
`timescale 1ns /1ps
module pixel_gen(
input clk_d,          // pixel clock
input [9:0] pixel_x,
input [9:0] pixel_y,
input video_on,
output reg [3:0] red=0,
output reg [3:0] green=0,
output reg [3:0] blue=0
);

always @(posedge clk_d)
begin
    if ((pixel_x ==0) || (pixel_x ==639) || (pixel_y ==0) ||
        (pixel_y==479))

        begin
            red <= 4'hF; green <= 4'hF; blue <= 4'hF;
        end

    else
        begin
            red <= video_on?(pixel_y>240? 4'hF:4'h0):(4'h0);
            green <= video_on?(pixel_y >240?4'h0:4'hF):(4'h0);
            blue <= 4'h0;
        end
    end
endmodule
```

Figure 11.6

This code generates horizontal stripes of green and red color on the screen (refer Figure 11. 8) with fine border of white color.

11.3.3. VGA Controller Complete Circuit

This task we will connect all the modules designed in Lab 9 and 10 and test their functionality on VGA monitor. Figure 11.7 represents the Top Level Module of the VGA controller.

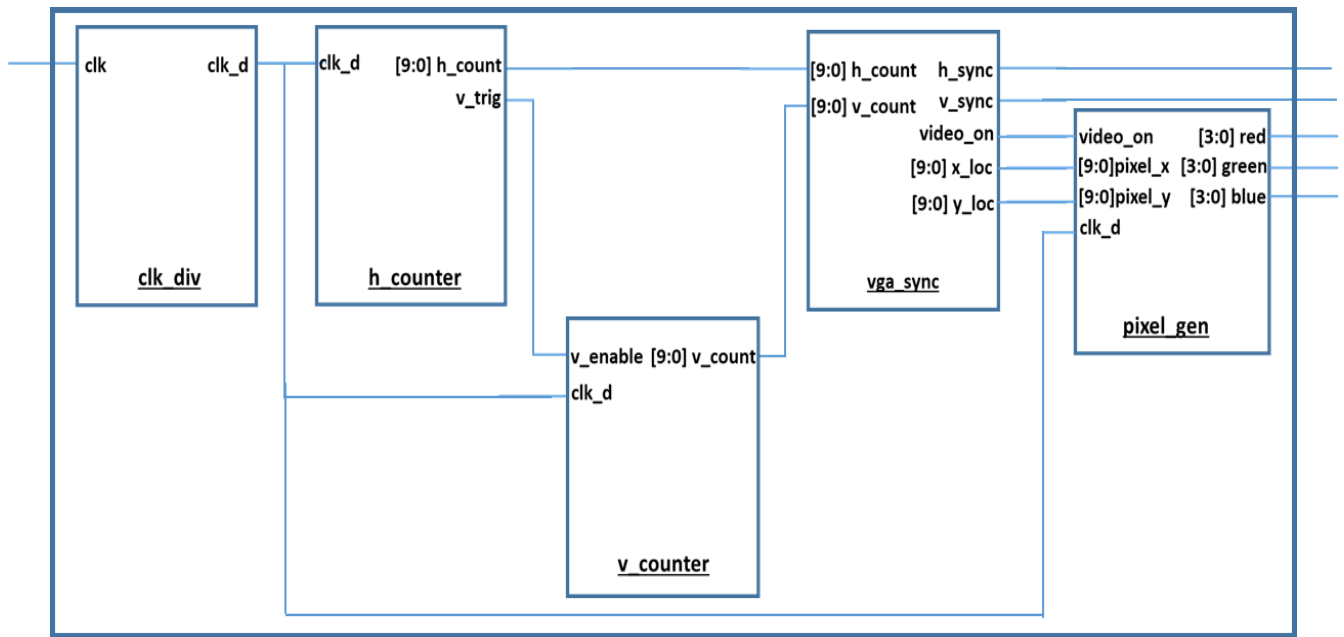


Figure 11. 7

Task c) VGA Controller

- Create a Top Level Module, as in Figure 11. 7. This module will receive a 100Mhz Clock as input and generate 1-bit h_sync and v_sync signals and three 4-bit signals for red, green and blue color.
- Implement¹ the VGA controller on BASYS-03 board and show output on VGA monitor. The display output is shown in Figure 11. 8. The VGA port connections are shown in Figure 11. 9. *Demonstrate your working VGA monitor to the respective RA.*

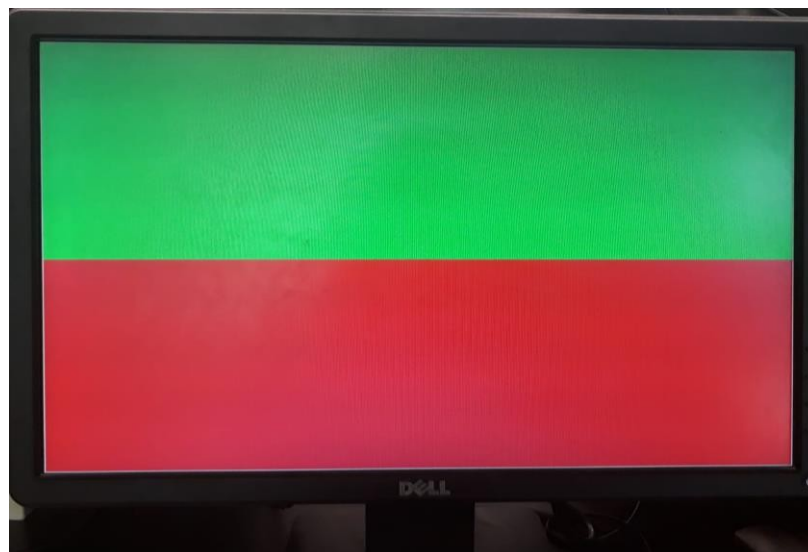


Figure 11. 8: VGA display

¹ Add your design files to Xilinx Vivado. Referring [Figure 11.9](#), create constraint file (.xdc file). For input clock signal, use W5 pin.

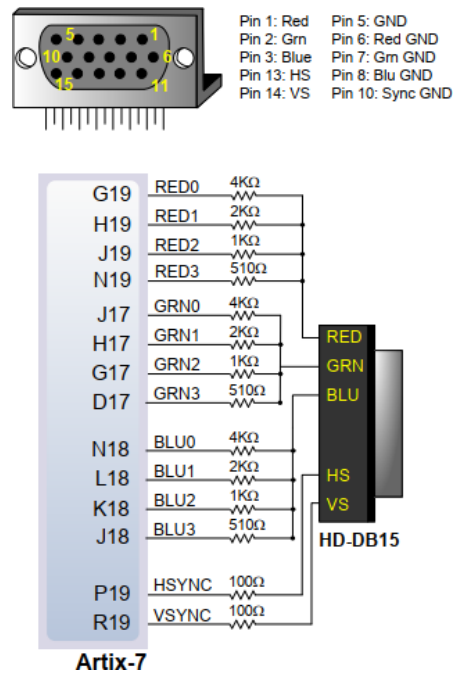


Figure 11. 9: VGA port connections

11.4. Exercise

Modify *pixel_gen* module to show an 8x8 checker board on VGA monitor as shown in Figure 11. 10. Each box on checked board should be a square (i.e., of equal width and height) you can fill the remaining area of screen with color of your choice. (Refer Table 11. 1)

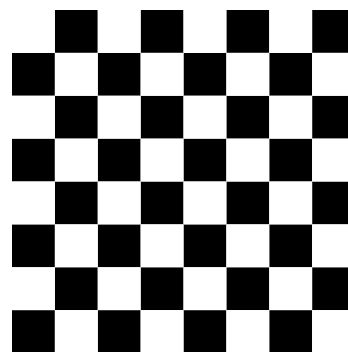


Figure 11. 10

Table 11. 1

Three-bit VGA color combinations			
Red (R)	Green (G)	Blue (B)	Resulting color
0000	0000	0000	black
0000	0000	1111	blue
0000	1111	0000	green
0000	1111	1111	cyan
1111	0000	0000	red



1111	0000	1111	magenta
1111	1111	0000	yellow
1111	1111	1111	white



Assessment Rubric

Lab 11

VGA Controller

Name:	Student ID:
-------	-------------

Marks Distribution:

	Task No.	LR2 Code	LR5 Results
In - Lab	Task a	/20	/05
	Task b	-	-
	Task c	/10	/15
Exercise		/20	/15
Total: 100		/50	/35
CLO Mapping		CLO3	CLO3

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission and Viva	/15	CLO 4

CLO	Total Points	Points Obtained
3	85	
4	15	
Total	100	

For description of different levels of the mapped rubrics, please refer to the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.



Lab Evaluation Assessment Rubrics

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR1	Circuit Layout	Connections between circuit components are mostly wrong. Circuit layout is cluttered. Needs guidance to make correct equipment/ component connections.	Few of the connections made between circuit components are wrong. Circuit layout is not neat and clean as per standards.	Circuit layout and connections are correct but not all connections are neat and clean as per standards.	Neat, clean and correct connections of all the circuit components/ equipment are made as per standard circuit diagram.
LR3	Troubleshooting	Unable to identify the fault/minimal effort shown in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR7	Viva	Response shows a complete lack of understanding of the assigned / completed task	Response shows shallow understanding of the assigned task	Response shows substantial understanding of the assigned task	Response shows complete understanding of the completed task. The student is able to explain all the related concepts.
LR10	Analysis	Results are not interpreted or are analyzed incorrectly.	Analysis of the obtained results is incomplete. Conclusions are not drawn.	Results are analyzed and concluded but are not well explained and justification is not provided with the help of reasoning.	Results are well analyzed supported with reasons. Good comparison is made between the theoretical and experimental results with correct and useful conclusion.
LR11	Design	Proposed design is unsubstantiated or does not satisfy most	Justification for proposed design is weak, and it only	Proposed design is substantiated, preferably through	Proposed design is substantiated, preferably through



		of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.
--	--	--	--	--	--